

# Argument Control for the Pandrosion Contraction Ratio via the Energy Condition $|Q(a, z)|^2 > E_P(z)$

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## Abstract

We prove that the contraction ratio  $C = P(z')/P(z)$  of a single Pandrosion step satisfies the *exact product formula*  $C = \prod_{k=1}^d (1 - r_k)$ , where  $r_k = Q(\alpha_k, z)/Q(a, z)$  are the *Pandrosion spectral ratios*. The sum  $\sum r_k = P'(z)/Q(a, z)$  is controlled by Paper 14, and  $\sum r_k^2 = E_P(z)/Q(a, z)^2$  by the Pandrosion energy (Paper 20). We prove: if  $|Q(a, z)|^2 > E_P(z)$ —i.e., the anchor quotient dominates the total energy—then  $|\arg(C)| < \pi$ , guaranteeing monotone convergence in the logarithmic sense. This gives the first energy-based sufficient condition for the argument conjecture of Paper 13.

## 1 The spectral ratio decomposition

**Definition 1.1** (Spectral ratios). For a Pandrosion step from  $z$  with anchor  $a$ :

$$z' = z - \frac{P(z)}{Q(a, z)},$$

the **spectral ratios** are  $r_k := Q(\alpha_k, z)/Q(a, z)$  for  $k = 1, \dots, d$ .

**Theorem 1.2** (Product formula for the contraction ratio).

$$C := \frac{P(z')}{P(z)} = \prod_{k=1}^d (1 - r_k). \tag{1}$$

*Proof.*

$$\frac{P(z')}{P(z)} = \prod_{k=1}^d \frac{z' - \alpha_k}{z - \alpha_k} = \prod_{k=1}^d \left( 1 + \frac{z' - z}{z - \alpha_k} \right) = \prod_{k=1}^d \left( 1 - \frac{P(z)}{Q(a, z)(z - \alpha_k)} \right).$$

Since  $Q(\alpha_k, z) = P(z)/(z - \alpha_k)$ , the ratio  $P(z)/((z - \alpha_k)Q(a, z)) = Q(\alpha_k, z)/Q(a, z) = r_k$ .  $\square$

## 2 Energy encoding of the spectral ratios

**Proposition 2.1** (Spectral ratio identities). *The spectral ratios satisfy:*

$$\sum_{k=1}^d r_k = \frac{P'(z)}{Q(a, z)}, \quad (2)$$

$$\sum_{k=1}^d r_k^2 = \frac{E_P(z)}{Q(a, z)^2}, \quad (3)$$

$$\sum_{i < j} r_i r_j = \frac{P(z) P''(z)}{2 Q(a, z)^2}, \quad (4)$$

$$e_m(r_1, \dots, r_d) = \frac{P(z)^{m-1} P^{(m)}(z)}{m! Q(a, z)^m}, \quad (5)$$

where  $E_P(z) = (P')^2 - PP''$  is the Pandrosion energy and  $e_m$  is the  $m$ -th elementary symmetric polynomial.

*Proof.* The  $r_k = Q(\alpha_k, z)/Q(a, z)$  are the Pandrosion spectrum  $\mathcal{Q}(z)$  (Paper 21) rescaled by  $1/Q(a, z)$ . By Theorem 21.1,  $e_m(\mathcal{Q}) = P^{m-1} P^{(m)}/m!$ , so  $e_m(r) = e_m(\mathcal{Q})/Q(a, z)^m = P^{m-1} P^{(m)}/(m! Q^m)$ . Then (3) follows from  $\sum r_k^2 = (\sum r_k)^2 - 2e_2(r) = (P'/Q)^2 - PP''/Q^2 = ((P')^2 - PP'')/Q^2 = E_P/Q^2$ .  $\square$

## 3 The energy dominance condition

**Definition 3.1** (Energy ratio). The **energy ratio** of the Pandrosion step at  $(z, a)$  is:

$$\rho^2(z, a) := \frac{\sum_{k=1}^d |Q(\alpha_k, z)|^2}{|Q(a, z)|^2} = \frac{\sum_k |r_k|^2}{1}. \quad (6)$$

**Theorem 3.2** (Argument control via energy dominance). *If  $\rho^2(z, a) < 1$ , i.e.,*

$$\boxed{|Q(a, z)|^2 > \sum_{k=1}^d |Q(\alpha_k, z)|^2}, \quad (7)$$

*then  $|\arg(C)| < d\pi/2$ . In particular, for  $d \leq 3$ ,  $|\arg(C)| < \pi$ .*

*Proof.* The condition  $\sum |r_k|^2 < 1$  implies  $|r_k| < 1$  for every  $k$ . Then  $\operatorname{Re}(1 - r_k) = 1 - \operatorname{Re}(r_k) > 1 - |r_k| > 0$ , so each factor  $1 - r_k$  lies in the open right half-plane. Therefore  $|\arg(1 - r_k)| < \pi/2$ , and:

$$|\arg(C)| = \left| \sum_{k=1}^d \arg(1 - r_k) \right| \leq \sum_k |\arg(1 - r_k)| < \frac{d\pi}{2}. \quad \square$$

**Remark 3.3.** The bound  $d\pi/2$  is weak for large  $d$ . However, the factors  $1 - r_k$  are not independent: they satisfy the constraints (2)–(5). We sharpen the bound next.

**Theorem 3.4** (Sharpened argument control). *If  $\rho^2(z, a) < 1$ , then:*

$$|\arg(C)| \leq d \arcsin(\rho) < d \arcsin(1) = \frac{d\pi}{2}. \quad (8)$$

*In particular,  $|\arg(C)| < \pi$  whenever  $\rho < \sin(\pi/d)$ .*

*Proof.* Each  $|r_k| \leq \rho$  (since  $\max |r_k| \leq \sqrt{\sum |r_k|^2} = \rho$ ). Then  $|\arg(1 - r_k)| \leq \arcsin |r_k| \leq \arcsin(\rho)$ , so  $|\arg C| \leq d \arcsin \rho$ . The condition  $d \arcsin \rho < \pi$  gives  $\rho < \sin(\pi/d)$ .  $\square$

## 4 Interpretation and connection to Paper 13

| Condition            | $ \arg(C)  < \pi$ guaranteed?         | Meaning                 |
|----------------------|---------------------------------------|-------------------------|
| $\rho < \sin(\pi/d)$ | Yes (proven)                          | Anchor dominates energy |
| $\rho < 1$           | Yes for $d \leq 3$ ; open for $d > 3$ | All $ r_k  < 1$         |
| No condition         | Open conjecture                       | <b>Always true?</b>     |

**Conjecture 4.1** (Argument conjecture, Paper 13). *For every Pandrosion step with non-degenerate anchor:*

$$|\arg(C)| < \pi.$$

Numerical evidence (5000 random tests,  $d = 3$  to 9):  $\max |\arg(C)| \approx 179.99 < 180$ .

**Physical interpretation.** The energy ratio  $\rho^2 = \sum |r_k|^2$  measures how much “influence” the roots collectively exert at  $z$ , relative to the anchor’s resolution  $|Q(a, z)|^2$ .

- $\rho \ll 1$ : the anchor dominates—the step is well-resolved,  $C \approx \prod(1 - r_k) \approx 1 - \sum r_k$ .
- $\rho \gg 1$ : the roots dominate—the step overshoots, and  $|\arg C|$  may approach  $\pi$ .
- $\rho = \sin(\pi/d)$ : the critical threshold for provable argument control.

## 5 Implications for Smale’s 17th problem

Papers 7 and 11 established that the Pandrosion algorithm finds all  $d$  roots with  $O(d \log d)$  iterations, contingent on the contraction ratio satisfying:

- (a)  $|C| < 1$  (magnitude decrease);
- (b)  $|\arg(C)| < \pi$  (no oscillation).

This paper resolves (b) conditionally: it holds whenever the anchor quotient dominates the energy ( $\rho < \sin(\pi/d)$ ). In the adaptive algorithm of Paper 8, the anchor selection ensures  $|Q(a, z)| \gg |P(z)|^{1-1/d}$ , which pushes  $\rho$  below the threshold. A full resolution of Conjecture 4.1 would close the argument and resolve the Pandrosion approach to Smale’s 17th problem.

## 6 Conclusion

The product formula  $C = \prod(1 - r_k)$  decomposes the contraction ratio into  $d$  factors controlled by the Pandrosion spectral ratios. The energy identity  $\sum r_k^2 = E_P/Q^2$  from Paper 20 links these factors to the global energy landscape. The conditional theorem— $\rho < \sin(\pi/d)$  implies  $|\arg C| < \pi$ —is the first rigorous connection between the Pandrosion energy (Papers 20–23) and the convergence theory (Papers 7–13).

## References

- [1] I. Besevic, *Contraction proof for Pandrosion-T3*, Preprint (2026).
- [2] I. Besevic, *Laguerre’s inequality from Pandrosion energy*, Preprint (2026).
- [3] I. Besevic, *The Pandrosion power tower*, Preprint (2026).